RESEARCH ARTICLE

Social hierarchy reveals thermoregulatory trade-offs in response to repeated stressors

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ABSTRACT

Coping with stressors can require substantial energetic investment, and when resources are limited, such investment can preclude simultaneous expenditure on other biological processes. Among endotherms, energetic demands of thermoregulation can also be immense, yet our understanding of whether a stress response is sufficient to induce changes in thermoregulatory investment is limited. Using the black-capped chickadee as a model species, we tested a hypothesis that stress-induced changes in surface temperature (T_s), a well-documented phenomenon across vertebrates, stem from trade-offs between thermoregulation and stress responsiveness. Because social subordination is known to constrain access to resources in this species, we predicted that T_s and dry heat loss of social subordinates, but not social dominants, would fall under stress exposure at low ambient temperatures (T_a), and rise under stress exposure at high T_a, thus permitting a reduction in total energetic expenditure toward thermoregulation. To test our predictions, we exposed four social groups of chickadees to repeated stressors and control conditions across a T_a gradient (n=30 days/treatment/group), whilst remotely monitoring social interactions and T_s. Supporting our hypothesis, we show that: (1) social subordinates (n=12), who fed less than social dominants and alone experienced stress-induced mass-loss, displayed significantly larger changes in T_s following stress exposure than social dominants (n=8), and (2) stress-induced changes in T_s significantly increased heat conservation at low T_a and heat dissipation at high T_a among social subordinates alone. These results suggest that chickadees adjust their thermoregulatory strategies during stress exposure when resources are limited by ecologically relevant processes.

KEY WORDS: Dominance, Stress, Thermoregulation, Birds

INTRODUCTION

The vertebrate stress response is characterised by coordinated behavioural and physiological modifications that enhance an individual's ability to cope with perceived challenges, or 'stressors'. Over the past century, research pertaining to stressor-induced physiological modifications has undergone overwhelming expansion (e.g. Cannon, 1932; Romero et al., 2009; Selye, 1950; lay literature: Sapolsky, 2004). This expansion has permitted the identification of numerous adaptations among vertebrates that not only serve to mediate stressors acutely but also serve to minimise the potential of wear and tear acquired from long-term activation of emergency pathways (that is, to reduce allostatic load; McEwen, 1998). To ecologists, adaptations of this latter form are of particular interest as they commonly involve re-allocation of resources among biological processes (e.g. 'trade-offs') that can reveal hierarchies of investment in a species. To date, the vast majority of studies investigating stress-induced trade-offs have focused on changes to reproduction (Calisi et al., 2008; Grachev et al., 2013; Kinsey-Jones et al., 2009; Kirby et al., 2008) and immune function (Stier et al., 2009; Svensson et al., 1998), probably owing to the importance of each process in dictating individual fitness, and the relatively high costs associated with their activation and maintenance (Martin et al., 2003; Ots et al., 2001; Speakman, 2008; Vleck, 1981; but see Merlo et al., 2014). Among endothermic vertebrates, however, the energetic costs of maintaining a balanced core temperature can often overwhelm those of both reproductive and immune function (Burness et al., 2010; King and Swanson, 2013; Nord et al., 2010), and failing to maintain a balanced core temperature can be equally as detrimental to individual fitness as failing to maintain reproduction or immunity. Nevertheless, trade-offs with respect to thermoregulatory investment in the presence of stressors remain poorly understood.

Changes in body temperature in response to stressors have been widely reported across vertebrate taxa (e.g. squamates: Cabanac and Gosselin, 1993; birds: Greenacre and Lusby, 2004; fish: Rey et al., 2015; old-world primates: Parr and Hopkins, 2000; rodents: Dymond and Fewell, 1998; Yokoi, 1966) and have been documented in medical literature for nearly two thousand years (Galen, 2nd century CE; Yeo, 2005). Although the proximate mechanisms driving thermal responses to stressors have since been well characterized (e.g. redistribution of blood flow, changes in brown adipose metabolism, secretion of pyrogenic cytokines; Kataoka et al., 2014; reviewed in Oka et al., 2001), an adaptive value of this phenomenon remains contended. Indeed, whilst some studies posit that thermal fluctuations per se may endow an individual with immunological advantages during coping (Oka et al., 2001), others argue that stress-induced thermal fluctuations are merely 'spandrels' (i.e. phenotypic traits that emerge, simply, as by-products of other adaptive phenotypes; Gould and Lewontin, 1979) that emerge from the redistribution of blood flow for other adaptive purposes (e.g. haemorrhage mitigation or re-distribution of oxygen and glucose; Bartlett, 1912; Jerem et al., 2015; Jerem et al., 2018), or as artifacts of experimental design (Andreasson et al., 2019; Nord and Folkow, 2019).

Recently, we proposed that an adaptive value of stress-induced changes in body temperature may be understood when contextualised according to an individual's perceived thermoregulatory costs, and when changes in body temperature are observed at the level of an individual's surface tissues

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